

MAHILINY JEGATHEESWARAN

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EDUCATION

Bachelor of Science in Electrical Engineering

University of Moratuwa

GPA: 3.80/4.2

Jan 2020 – Jul 2024

Sri Lanka

SKILLS

Programming Languages: Java, Python, JavaScript, HTML

Frameworks & Libraries: Spring, Spring Boot, Node.js, YOLOv5, OpenCV, TensorFlow, PyTorch

Databases: SQL, PostgreSQL, DynamoDB, Redis

Cloud & Tools: AWS (S3, EC2, Cognito, Lambda), Git, Bitbucket, Kafka, Jenkins (CI/CD)

Data Science & ML: Machine Learning, Computer Vision, Web Scraping

EXPERIENCE

Software Engineer

Smartzi

Jul 2024 - Present

Colombo, Sri Lanka

- Designed and implemented scalable microservices and RESTful APIs using Java and Spring Boot Framework to power Smartzi's driver app, customer app, control panel, and B2B portal.
- Improved backend performance and maintainability by optimizing database interactions through PostgreSQL stored procedures and efficient query design.
- Designed and implemented microservices architecture using Spring Boot and Spring Security with OAuth2, including Cognito-based security integration.
- Managed production deployments, conducted application monitoring, and resolved production-level issues and defects.

Data Engineer - Part time

Smartzi

Jul 2023 - jun 2024

Colombo, Sri Lanka

- Created job bundling mechanisms to group and manage scheduled tasks, reducing resource usage and execution time.
- Built custom web scrapers using Python to extract structured data from websites for analysis and reporting.

Data Science Intern

Smartzi

Jan 2023 - jun 2023

Colombo, Sri Lanka

- Contributed to diverse backend projects including OCR-based UK license verification and dynamic vehicle group reconfiguration.
- Utilized Python, PostgreSQL, and Tesseract OCR for document parsing and validation workflows.

PROJECTS

Smartzi - Marketplace

Java 21, Spring Boot, SQL, PostgreSQL, AWS, Redis, Kafka

Jan 2025 - Present

Design and implement an intelligent job assignment algorithm based on proximity, availability, price rate, GPS, and vehicle group. Enhance driver partner management features to track and qualify drivers for improved reliability.

Smartzi – World Transfer

Java 17, Spring Boot, AWS DynamoDB

Oct 2024 - Dec 2024

Development and implementation of a backend service to periodically retrieve and process client bookings (World Transfer B2b's bookings) from an external API, ensuring accurate and timely synchronization with Smartzi's internal systems.

Smartzi - Student Finance England (SFE)

Jul 2024 - Oct 2024

Java 8, Spring, SQL, PostgreSQL, AWS

Worked on backend development for Smartzi's B2B integration with Student Finance England (SFE), allowing the booking of rides and account management for students eligible for the Disabled Students' Allowance (DSA).

Smartzi - Job Bundling

Mar 2023 - Jun 2023

Python, SQL, H3, Joblib, Geospatial indexing

Build a Python algorithm for Smartzi to create job bundles by considering job start times and locations. The algorithm efficiently groups jobs based on proximity and timing, optimizing scheduling, and improving operational efficiency.

Intelligent Beach Cleaning Robot - Final Year Project

Aug 2023 - Jul 2024

Python, YOLOv5, OpenCV, CNN, Arduino, Computer Vision

Build a Python algorithm for Smartzi to create job bundles by considering job start times and locations. The algorithm efficiently groups jobs based on proximity and timing, optimizing scheduling, and improving operational efficiency.

AWARDS

- **IEEE R10 Robotics Competition (SL Section) 2024 – First Place.**
- **Dean's List**, University of Moratuwa - 2nd, 4th, 7th, and 8th semesters
- **Best Paper Award Competitors, Engineering Research Unit - University of Moratuwa**

PUBLICATIONS

- **Intelligent Beach Cleaning Robot with Dual Modes of Refuse Collection and YOLO based Detection** - 4th International Conference on Electrical Engineering (EECon) - 2024 [IEEE Xplore](#)
- **Design of Refuse Collection Mechanisms for an Intelligent Beach Cleaning Robot** - University of Moratuwa Engineering Research Unit (ERU) 2023 [here](#)

REFERENCES

- **Dr. V. Logeeshan**
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- **Prof. A.G. Buddhika P. Jayasekara**
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